Pre-reconstruction RSD B000-only validation | contract and status

Status: COMPLETE_WITH_DIAGNOSTICS | generated: 2026-10-02T04:39:32 UTC
Baseline summaries: 18/18; converged: 18/18
Sensitivity cases: 6; no missing result is treated as a pass.
Measured boxes: mesh256=800/800; mesh512=3/3.

Observable and data contract
Sugiyama B000(k1,k2), the same radial edges and 44 certified shell pairs as real space.
The closing third leg is unrestricted; excluded zero-mode terms keep the full-shell denominator.
kmax: .08 / .10 / .12 / .14 / .18 / .20; B points: 9 / 14 / 20 / 27 / 35 / 44.
The .15 cut has exactly the .14 indices and aliases that chain, rather than duplicating it.
500 disjoint fiducial covariance ICs (1000-1499); 100 validation ICs per PNG sign (0-99).
100-box mean data vector; unaveraged single-box covariance from Fid500; V=1

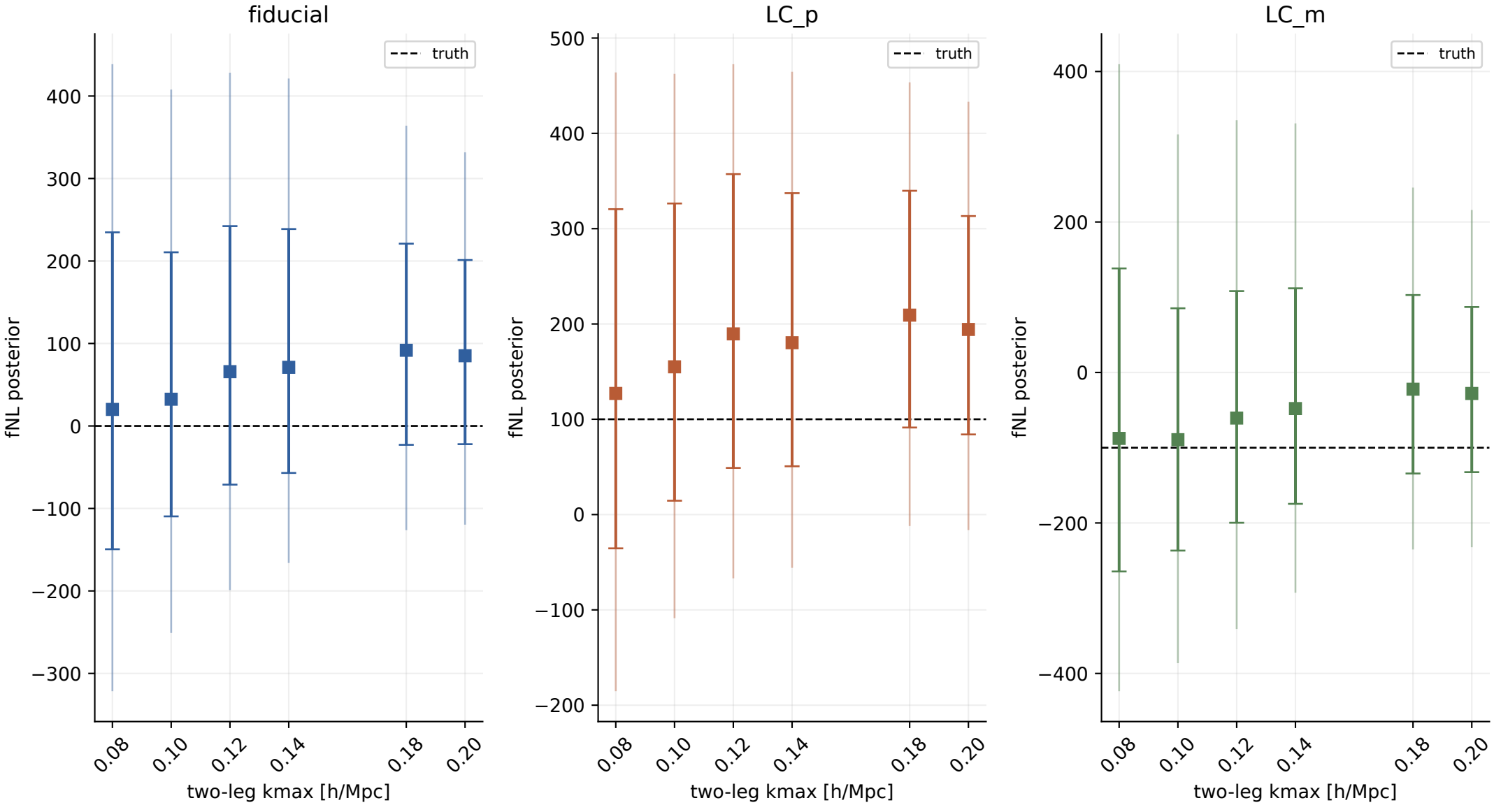
Model and inference
Pre-reconstruction tree G + fNL L + fNL^2 Q; fixed f=.7625790038 and external p=1.179.
Free b1 and nuisance biases/noise; no P data, P-derived b1 prior, IR, FoG, loops or counterterms.
Exact-lattice source FFT projection and independent native/direct-enumeration tests.
Four independent emcee ensembles, 32 walkers each; resumable HDF; budget-only extensions recorded separately.
Convergence gates: rank Rhat, bulk/tail ESS, length/tau, stationarity, ensemble spread, acceptance.

Scientific interpretation
A fit containing the true fNL once is not a validity certificate. Unconverged fits are excluded.
Mean target + Csingle is a single-box-precision model test, not the sampling likelihood of the mean.
GOF is only a forecast-weighted residual indicator, not a calibrated chi2 p-value for averaged data.
Finite-PNG stochastic closure, p uncertainty and covariance cosmology dependence remain limitations.
Numerical agreement does not validate reconstructed RSD data or full RSD IR resummation.

Contiguous conditional baseline range candidates
fiducial: kmax=0.2; boundary prior test=stable
LC_p: kmax=0.2; boundary prior test=stable
LC_m: kmax=0.2; boundary prior test=stable
Noise rank audit: 7 parameters; anisotropic shape retained=True

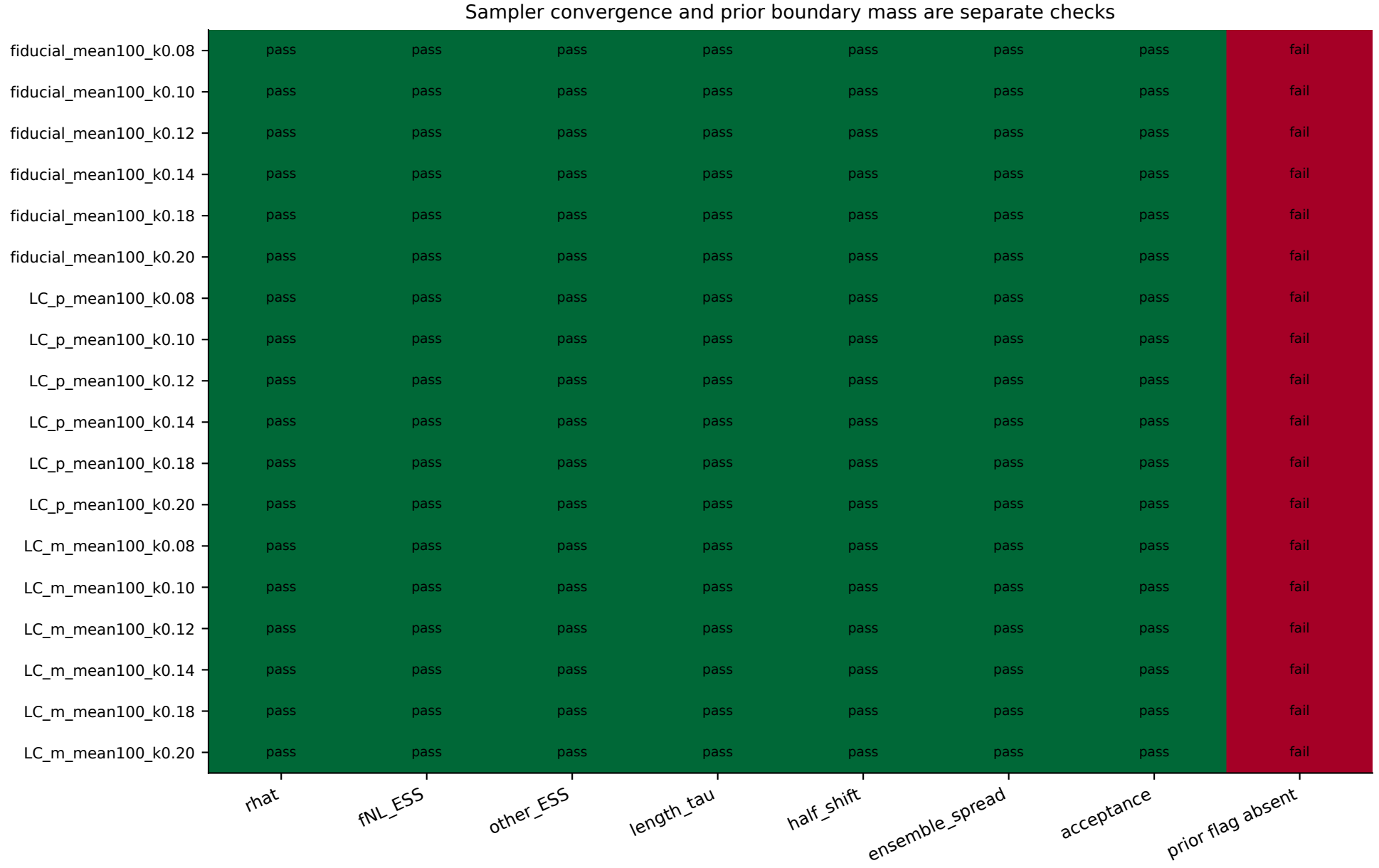
Execution provenance: CANCELLED_FOR_USER_REQUESTED_LOGIN_EXECUTION (IDs are not proof of completion)
baseline18: 59183525; sensitivity_freeze: 59183527; sensitivity_up_to6: 59183529; final_report: 59183533
Login runner: login12; CPU affinity [6, 7, 8, 9, 10, 11, 12, 13]; max 8 logical CPUs.

Pre-reconstruction RSD B000-only validation | mean100 cutoff scan



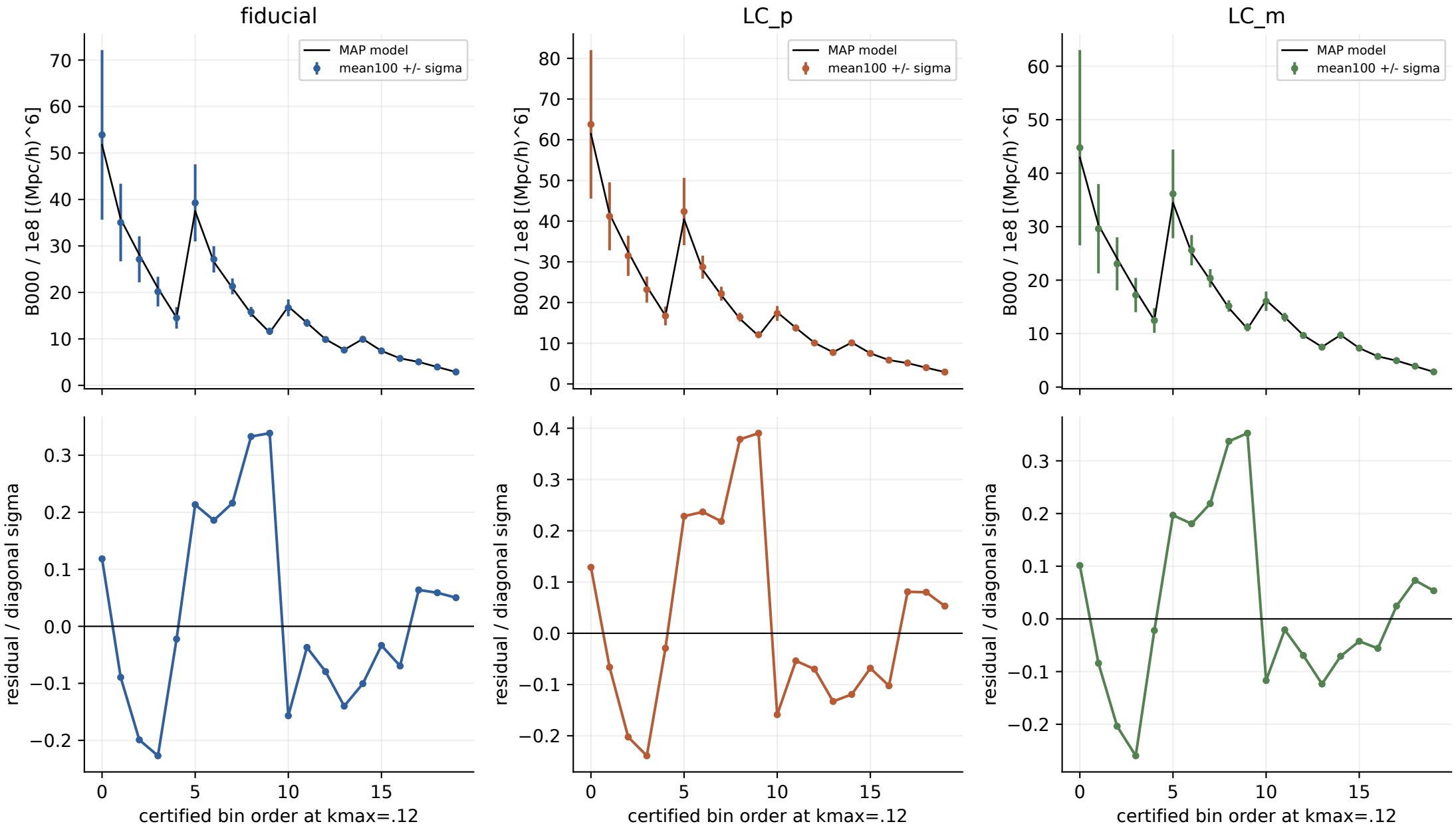
Intervals: 68% (thick), 95% (thin). Gray = not converged; square = nuisance/prior boundary flag.

Pre-reconstruction RSD B000-only validation | independent-chain diagnostics

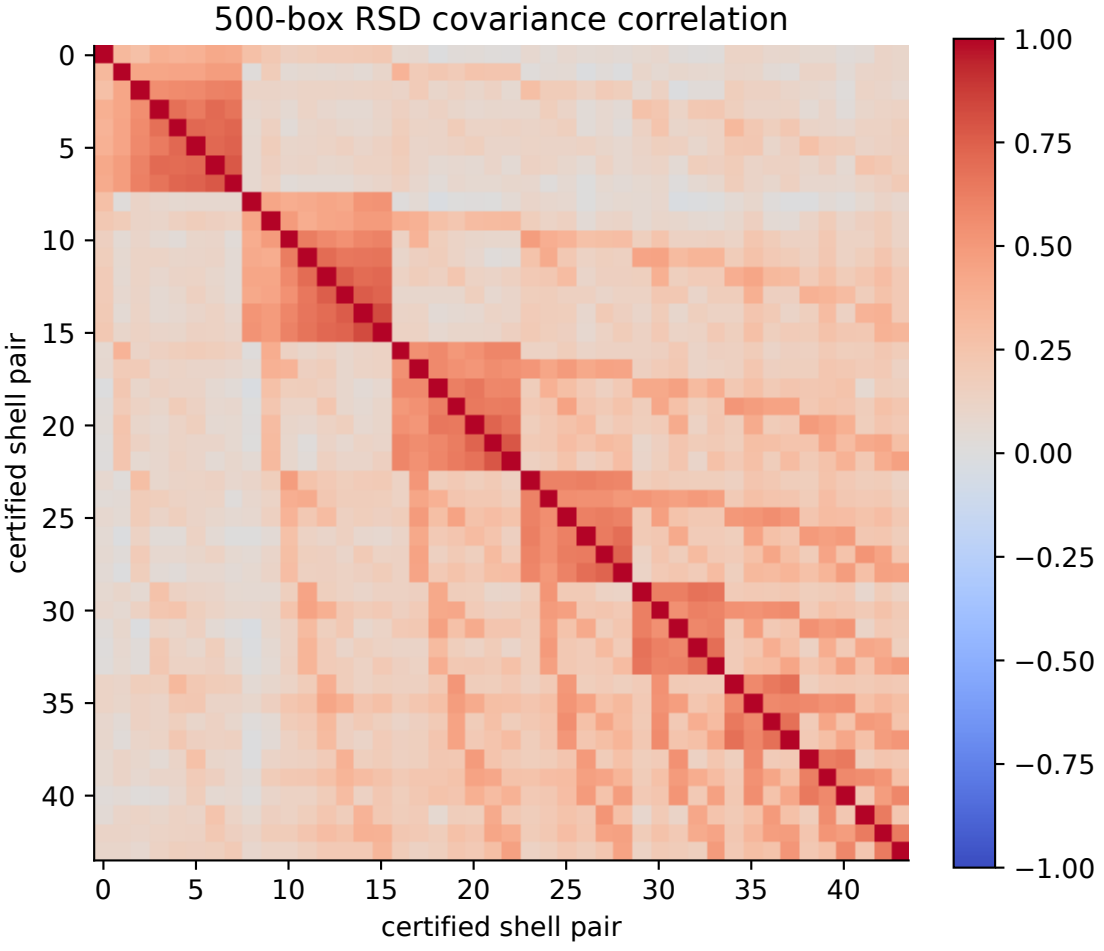


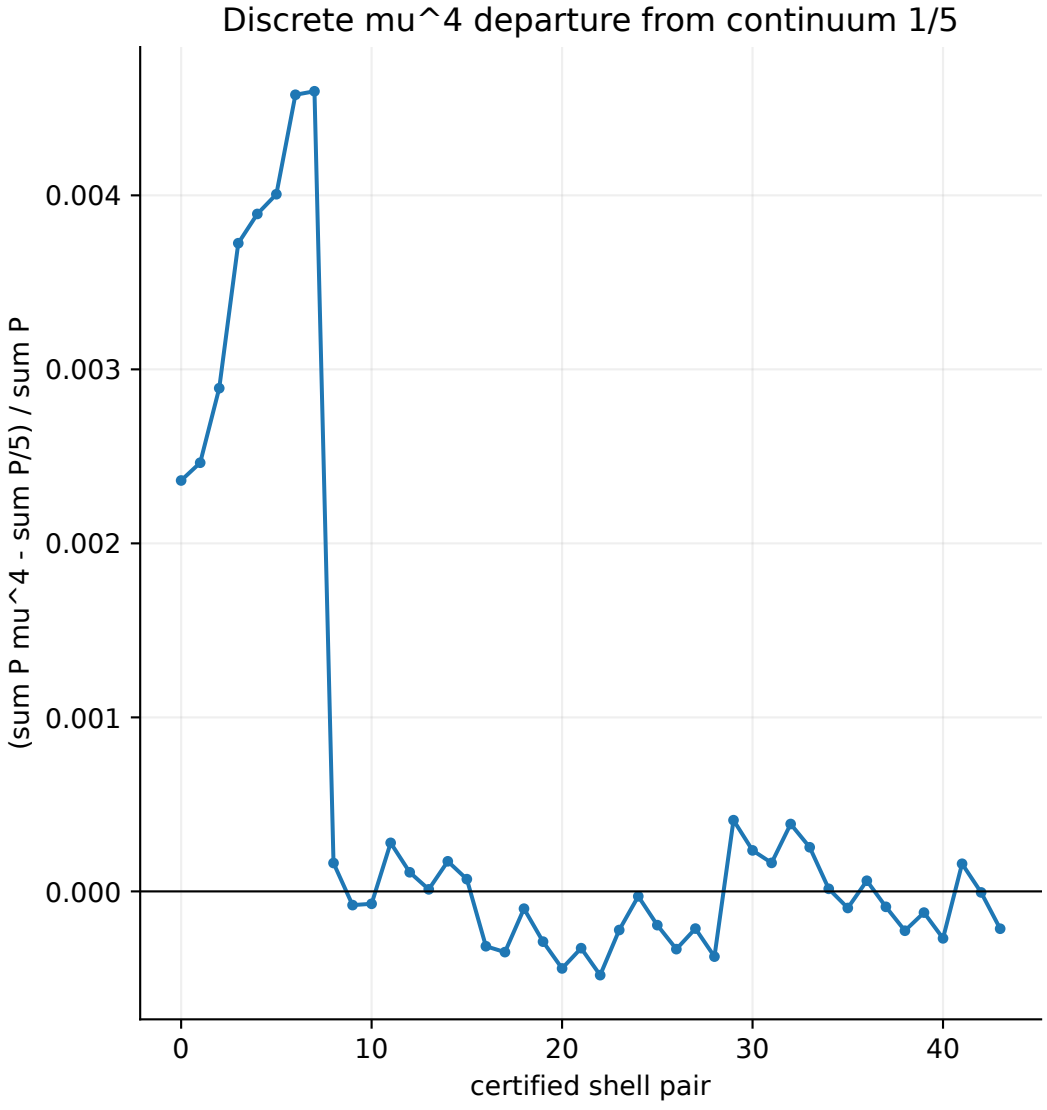
B-only; no P-derived prior. All validity statements are conditional.

Pre-reconstruction RSD B000-only validation | B-only fit and residuals



Pre-reconstruction RSD B000-only validation | covariance and noise rank

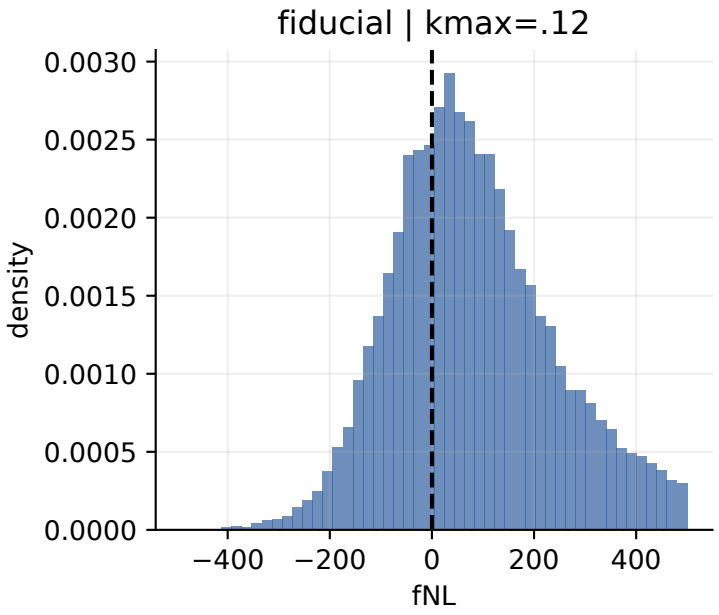


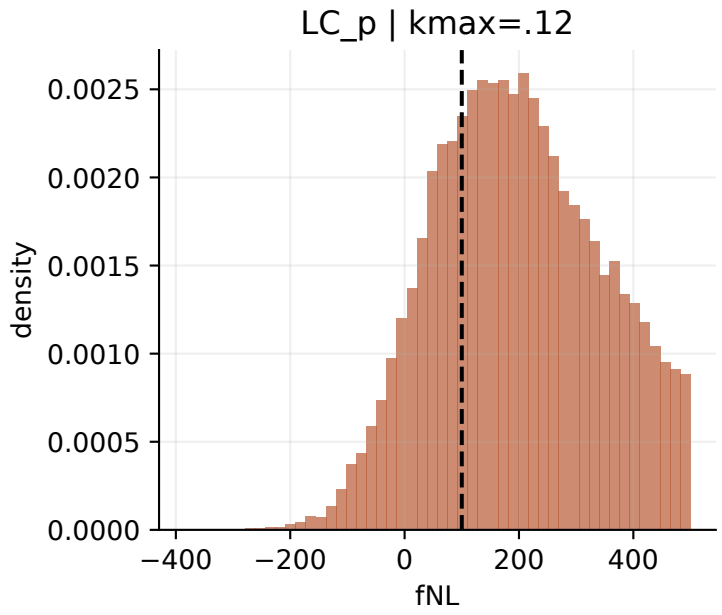


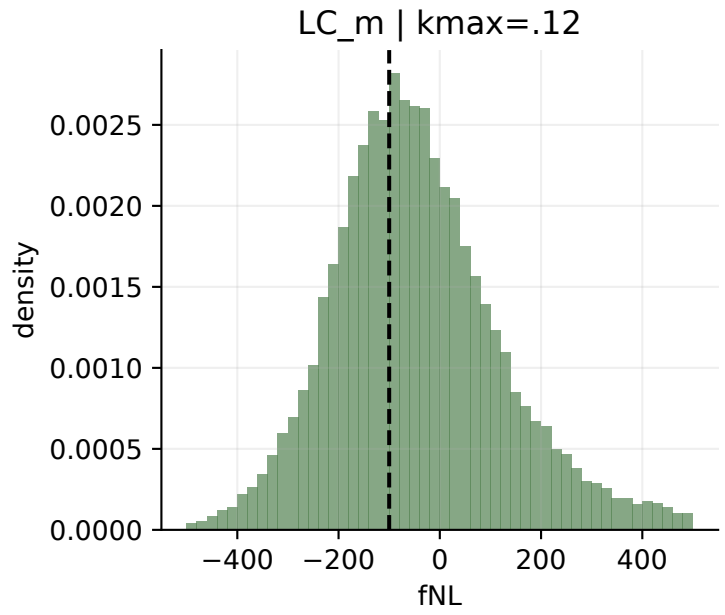
Noise audit: 7 parameters; retained independent SP shape: True

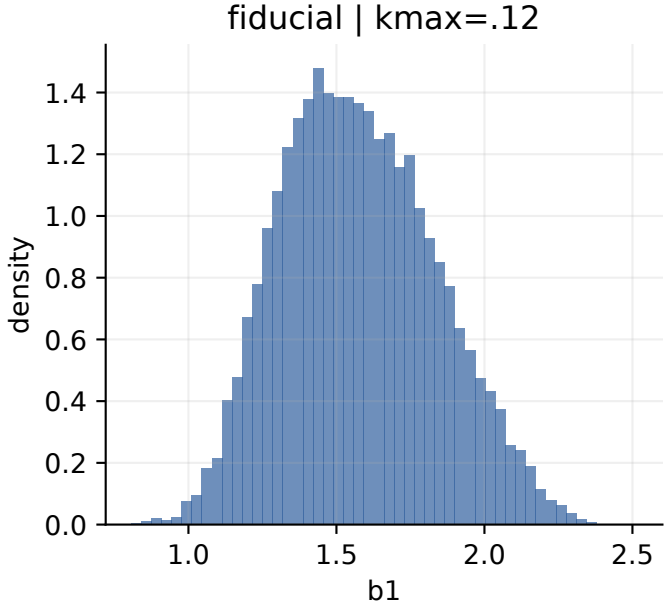
This Gaussian noise audit is not finite-PNG stochastic closure.

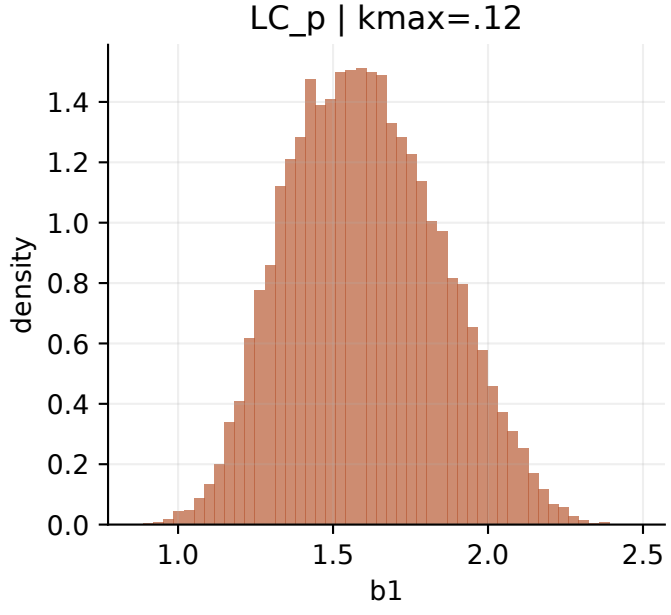
Pre-reconstruction RSD B000-only validation | selected marginalized posteriors

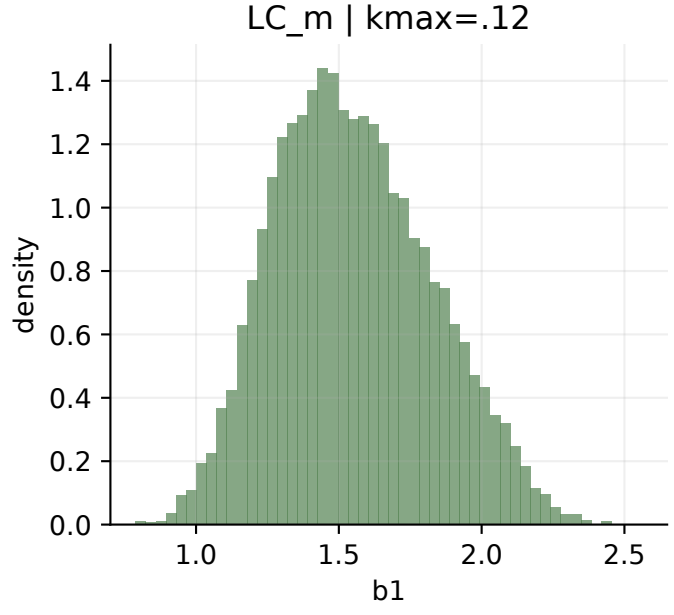






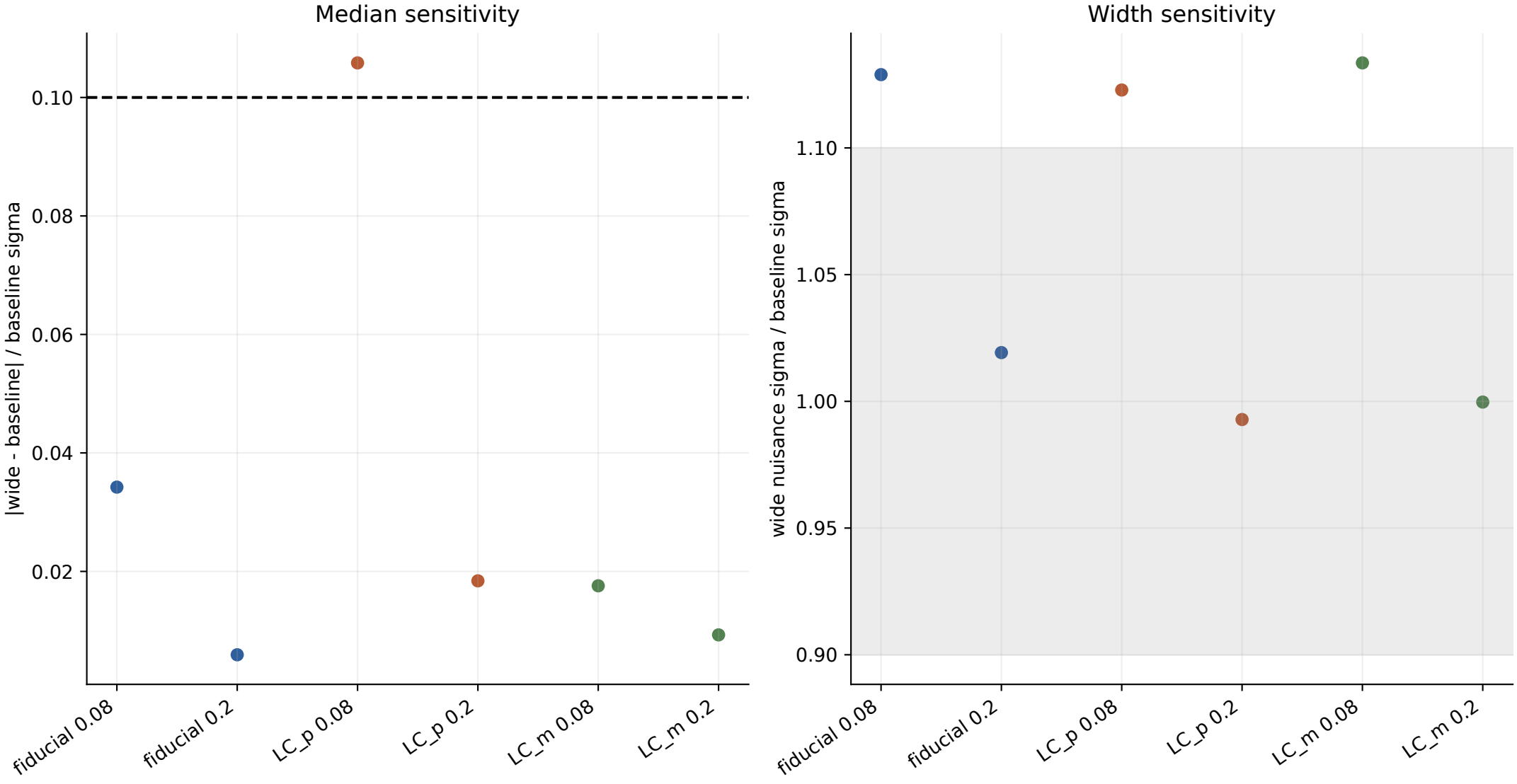






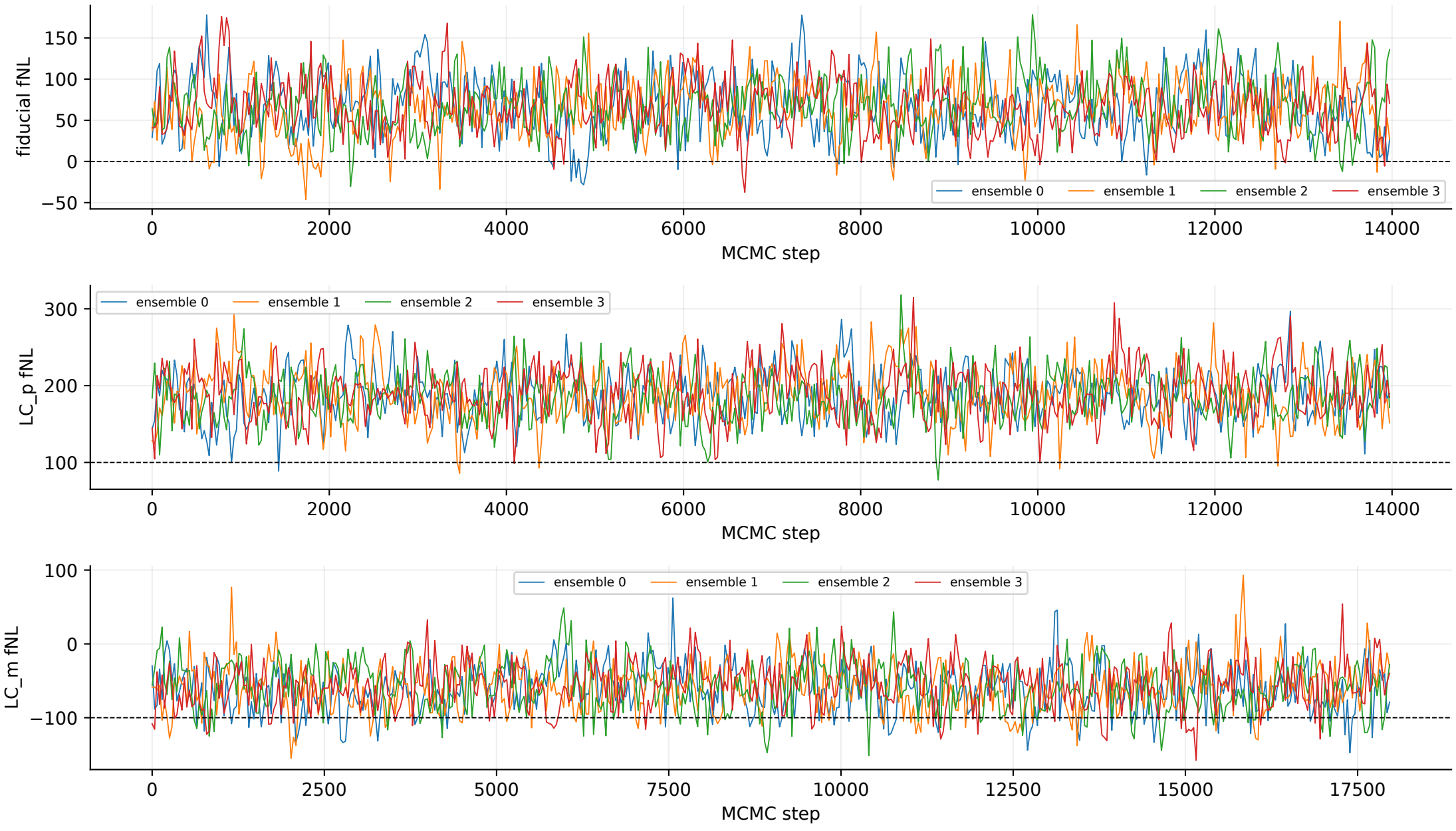
B-only; no P-derived prior. All validity statements are conditional.

Pre-reconstruction RSD B000-only validation | nuisance prior sensitivity



Only nuisance bounds widened x2; fNL, b1, p, data and covariance fixed. Both chains must converge.

Pre-reconstruction RSD B000-only validation | independent ensemble traces at .12



B-only; no P-derived prior. All validity statements are conditional.